

Project Progress Report on

AI and Computer Vision-Based Waterlogging Detection & Alert System

**Submitted in partial fulfilment of the requirement for the award of the
degree of**

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE & ENGINEERING

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Project Progress Report No: 2



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CANDIDATE'S DECLARATION

We hereby certify that the work which is being presented in the project progress report entitled “**AI and Computer Vision-Based Waterlogging Detection & Alert System**” in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science in the Department of Computer Science and Engineering of the Graphic Era (Deemed to be University), Dehradun shall be carried out by the undersigned under the supervision of Dr. Sarvesh Vishwakarma ,Professor Department of Computer Science and Engineering, Graphic Era (Deemed to be University), Dehradun.

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The above mentioned students shall be working under the supervision of the undersigned on the “**AI and Computer Vision-Based Waterlogging Detection & Alert System**”

Supervisor

Head of the Department

Examination

Name of the Examiners:

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Chapter 1

Introduction and Problem Statement

1.1 Introduction

Waterlogging has become a recurring challenge in both residential and urban environments. Rooftop drains often get blocked due to fallen leaves, dust, or debris, which leads to water accumulation during rainfall. If such situations go unnoticed, they may cause seepage, leakage, dampness, structural deterioration, and fungal growth inside buildings.

In urban settings, a similar issue is seen on streets and highways where clogged potholes or roadside drains lead to unwanted stagnation of water. This results in traffic disruptions, increased accident risks, rapid road degradation, and numerous safety concerns.

Traditional methods for waterlogging inspection rely heavily on manual checking or physical sensors. Manual inspections are slow and often detect the issue only after the damage has intensified. On the other hand, sensor-based systems require wiring, additional hardware, and maintenance, making them costly and less scalable.

To address this, our project proposes a **camera-based, AI-driven Waterlogging Detection and Alert System**, which utilizes Computer Vision and Deep Learning techniques to identify early signs of water accumulation. A simple CCTV or IP camera acts as the only hardware component, capturing continuous video frames. These frames are analyzed using a deep learning model—in our case, a modified InceptionV3 architecture—to classify whether waterlogging is present or not.

This solution removes the dependency on physical sensors, making the system fully wireless, non-intrusive, cost-effective, and easy to deploy. With features such as real-time monitoring, visual proof generation, and automated alerts, the system aims to provide a practical and intelligent approach to prevent water-related damage in buildings and urban infrastructure.

1.2 Problem Statement

Waterlogging poses a significant threat to rooftops, households, and road infrastructure. Accumulated water due to clogged drains can weaken building foundations, create leakage issues, and promote mosquito breeding. On roads, water stagnation leads to unsafe driving conditions and accelerates surface degradation.

Existing detection approaches—manual inspection and sensor-based systems—are either inefficient, expensive, or require continuous maintenance. Manual methods detect problems late, while sensor-based setups involve complex installation and frequent repairs.

There is a strong need for an automated, real-time, and low-cost waterlogging detection system that:

- relies only on camera feeds,
- eliminates the need for physical sensors,
- provides early alerts to prevent damage,
- is scalable for both household rooftops and large urban areas.

Our project aims to fill this gap with a Computer Vision-based approach that intelligently detects waterlogging using deep learning models and generates timely alerts along with evidence.

Chapter 2

Objectives

The primary goal of this project is to develop a smart, camera-based, AI-driven system that continuously monitors rooftops, drains, and urban roads for waterlogging or blockages, and provides timely alerts to homeowners or authorities. Here are the detailed objectives:

- **Enhancing Urban and Household Safety:** The system aims to improve safety and prevent structural damage caused by water accumulation. By using AI-based computer vision, it detects waterlogging early, reducing the risk of accidents, mosquito breeding, and infrastructure deterioration. The research goal is to create a system that proactively prevents water-related hazards in households and urban areas.
- **Real-Time Monitoring and Automated Alerting:** A key objective is to implement real-time monitoring that triggers immediate alerts through wireless notifications (SMS, email, or push notifications) when waterlogging is detected. This minimizes human intervention, enabling quick response and timely maintenance. The system should also provide visual evidence, such as images or short video clips, to confirm the alerts.
- **Wireless and Non-Intrusive Solution:** Unlike traditional waterlogging detection systems that rely on IoT devices or physical sensors, this project focuses on a completely wireless solution using cameras only. The objective is to design a nonintrusive system that is easy to install, cost-effective, and scalable across households, highways, and urban infrastructure.

Chapter 3

Project Work Carried Out

1. Live Image Capture

- Implemented a live image capture module using a smartphone as an IP camera(IP Webcam app).
- Accessed the latest camera frame via HTTP using a snapshot URL of the form `http://<phone_ip>:8080/shot.jpg`.
- Ensured both phone and PC were on the same Wi-Fi so the PC could reach the phone's private IP.
- Developed a Python script using the requests library to periodically send HTTP GET requests and fetch the current frame.
- Stored each received JPEG image in a "snippets" directory with timestamp-based filenames to maintain time order.

2. Data Preprocessing

- To ensure compatibility with the deep learning model:
- All images were resized to $224 \times 224 \times 3$ resolution.
- Pixel values were normalized.
- Data augmentation techniques such as rotation, zooming, brightness variation, shifting, and horizontal flipping were applied to avoid overfitting.
- The dataset was split into 70% training, 15% validation, and 15% testing.

3. Model Selection and Architecture

- After evaluating multiple deep learning architectures such as VGG16, ResNet50, and MobileNet, InceptionV3 was selected because:
 - it performs well with transfer learning,
 - it extracts multi-scale features,
 - it requires fewer parameters compared to very deep networks,

- and it gives strong accuracy on small custom datasets.

4. Model Modification

- The pre-trained InceptionV3 model was customized by:
- adding Global Average Pooling layer,
- adding a Dropout layer to reduce overfitting,
- adding a Dense output layer with sigmoid activation for binary classification,
- fine-tuning the last few layers for drainage-specific features.
- Training and Optimization
- The model was trained for 20 epochs using the Adam optimizer (learning rate: 0.001).
- Binary Cross Entropy was used as the loss function.
- Batch size was set to 32.
- The model showed high accuracy and stable learning behavior during training and validation.
- Initial Results
- The model demonstrated strong potential in correctly classifying waterlogged conditions.
- Predictions remained consistent across varied lighting, angles, and minor camera vibrations.
- The system is now ready for integration with real-time CCTV input.

Chapter 4

Future Work Plan

Sl. No.	Work Description	Duration in Week
1	Integration of trained model with live camera feed	1-2
2	Development of automated alert (SMS/Email) system	1-2
3	Creation of monitoring dashboard	2-3
4	Fine-tuning using additional collected images	3-4

Chapter 5

Weekly Task

Week No.	Date: From-To	Work Allocated	Work Completed (Yes/No)	Remarks	Guide Signature
1	1-10-25 to 18-10-25	Research and Planning	Yes		
2	20-10-25 to Present	Dataset Collection and Literature Review	In Progress		
3	20-10-25 to 5-11-25	Camera Setup for image capturing	Yes		
3	27-10-25 to 10-11-25	Model selection and custom architecture design	Yes		
4	13-12-25 to 03-02-26	Base Model Performance	Yes		
5		Fine Tuning and Model Performance Evaluation	No		
6	20-10-25 to Present	Research Paper Review	In Progress		

References

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